VIETNAM NATIONAL UNIVERSITY, HANOI

INTERNATIONAL SCHOOL

COURSE: INE 323401 PROJECTS IN FINANCIAL TECHNOLOGY

Machine Learning for Asset Clustering and Portfolio Optimization via the Fama-French Model

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Ha Noi, May 2026

Acknowledgments

Acknowledgments

I would like to express my sincere gratitude to Tran Vu Anh for the invaluable guidance and continuous support throughout the entire research process. From the early stages of defining the research scope to the final refinement of the methodology, the thoughtful feedback and constructive suggestions provided at every step were essential in shaping the direction and quality of this paper.

In particular, the advice on structuring the Walk-Forward Validation framework, selecting appropriate statistical tests, and interpreting the clustering results helped me approach each challenge in a more rigorous and systematic way. The patience and willingness to discuss ideas made a significant difference in how this research developed.

This paper would not have reached its current form without that dedication and support. I am truly grateful for the time and effort invested in guiding this project.

Abstract

This study designs, implements, and rigorously validates an automated quantitative portfolio management system integrating three analytical stages: (1) Fama-French three-factor OLS regression to extract five-dimensional risk feature vectors $[\alpha, \beta_{Mkt}, \beta_{SMB}, \beta_{HML}, \sigma_{\varepsilon}]$ for each security; (2) K-Means clustering (K = 3) with Centroid Sorting to group 15 S&P 500 equities into economically interpretable risk clusters based on latent factor exposures; and (3) SLSQP portfolio optimization with Ledoit-Wolf covariance shrinkage, per-asset weight caps (15%), and a cluster-level concentration constraint (40%). The pipeline is validated through Walk-Forward Validation (WFV) across 108 monthly rebalancing windows spanning 2015–2023, ensuring complete elimination of look-ahead bias and data leakage.

Out-of-sample results demonstrate that FF-KMeans-SLSQP achieves a Sharpe ratio of 0.6795, cumulative return of 234.35%, annualized return of 14.39%, and a Calmar ratio of 0.4169, leading or tying for best across 7 of 9 performance metrics. A central emergent finding is that K-Means, operating exclusively on Fama-French coefficients without explicit economic labels, independently rediscovers the Growth/Value dichotomy, confirming that the algorithm captures genuine cross-sectional economic structure from latent factor data.

Hypothesis H_1, that the strategy achieves a significantly higher Sharpe ratio than benchmarks, is confirmed against the SPY ETF passive benchmark (Lo (2002) z-test, p = 0.0001), but remains statistically inconclusive against quantitative benchmarks due to limited statistical power (36.5% at n = 2,263 daily observations). Hypothesis H_2 is partially supported: during the Federal Reserve tightening cycle (2022), the strategy records −5.59% versus −20.54% for SPY, with a near-threshold KS p-value of 0.0554. Hyperparameter sensitivity analysis across nine configurations confirms a Sharpe range of [0.5716, 0.8041], establishing baseline robustness.

Keywords: *Fama-French three-factor model, K-Means clustering, SLSQP optimization, Ledoit-Wolf shrinkage, Walk-Forward Validation, portfolio optimization, machine learning in finance.*

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Chapter I: Introduction

1.1 Background and Motivation

The management of investment portfolios in an increasingly complex global financial environment demands analytical frameworks that transcend the limitations of classical single-factor asset pricing theory. The Capital Asset Pricing Model (CAPM), formalized by Sharpe (1964) and Lintner (1965), established the foundational theoretical relationship between an asset's systematic market risk (beta) and its expected return. While CAPM's parsimony made it the dominant paradigm for several decades, a substantial body of empirical evidence accumulated against its adequacy as a complete description of cross-sectional return variation. Banz (1981) documented the size effect, indicating that small-capitalization equities earn returns exceeding CAPM predictions, while Rosenberg et al. (1985) established the value premium, wherein high book-to-market stocks systematically outperform low book-to-market stocks after adjusting for market beta. Fama and French (1992) synthesized these findings, demonstrating that beta alone provides minimal explanatory power once size and book-to-market characteristics are controlled for.

Fama and French (1993) formalized these empirical regularities in the three-factor model, augmenting the market factor with two mimicking portfolio returns: SMB (Small Minus Big), capturing the return differential between small- and large-capitalization firms, and HML (High Minus Low), capturing the premium associated with high versus low book-to-market firms. This multidimensional risk framework explains approximately 90–95% of time-series variation in diversified portfolio returns, substantially improving upon CAPM's 70–80% explanatory power.

The model has since become the standard benchmark for empirical asset pricing and performance evaluation in both academic and practitioner contexts.

Despite the theoretical richness of the Fama-French framework, its practical application to real-time portfolio construction faces substantial operational challenges. The high-dimensional and time-varying nature of factor exposures across multiple securities requires adaptive computational tools that respond dynamically to shifting market regimes. Furthermore, the classical mean-variance optimization framework of Markowitz (1952), while elegant in theory, suffers in practice from severe instability in parameter estimation. Merton (1980) demonstrated theoretically that historical mean return estimates are dominated by estimation noise, implying that optimization effectively targets minimum-variance rather than

true mean-variance efficiency, while Chopra and Ziemba (1993) showed quantitatively that errors in the expected return vector are roughly 11 times more costly than errors in the covariance matrix (for a medium-risk-aversion investor). This instability motivates both robust covariance estimation and the use of machine learning techniques to extract structural information beyond simple historical moments.

This study addresses these challenges by designing, implementing, and rigorously evaluating an automated quantitative portfolio management system that integrates K-Means machine learning clustering with the Fama-French three-factor framework and SLSQP optimization. By anchoring the clustering feature space in theoretically motivated Fama-French factor exposures, the system leverages economic theory to structure the machine learning problem, while the Walk-Forward Validation framework ensures that all reported performance estimates reflect genuine out-of-sample predictive ability.

1.2 Research Gap

The existing literature at the intersection of factor investing and machine learning, while growing rapidly, exhibits three critical gaps that this study addresses directly. First, while clustering algorithms have been applied to financial portfolio construction in several forms, notably Hierarchical Risk Parity (Lopez de Prado, 2016) and correlation-based groupings, the systematic integration of K-Means clustering operating directly on Fama-French regression feature vectors as a pre-optimization step for constrained portfolio construction remains underexplored. Most existing studies apply clustering either as an independent classification tool or outside the context of quantitative optimization.

Second, a substantial portion of the empirical portfolio optimization literature relies on simple hold-out testing or in-sample evaluation (De Miguel et al., 2009), which is susceptible to look-ahead bias and produces overly optimistic performance estimates. The rigorous application of Walk-Forward Validation, in which every stage of the pipeline, including StandardScaler normalization, OLS regression, and K-Means clustering, is re-fitted exclusively within each training window, represents a more demanding evaluation standard that few published undergraduate or practitioner studies implement fully.

Third, statistical testing of Sharpe ratio differences in the quantitative finance literature frequently relies on standard t-tests, which Lo (2002) demonstrated have severely limited power for detecting differential Sharpe ratios in daily return series due to the high variance of individual daily observations. The comprehensive application of Lo's (2002)

asymptotic z-test alongside formal power analysis, Newey-West HAC-adjusted standard errors, and bootstrap confidence intervals provides a methodological completeness that is rarely achieved in comparable capstone research. This study fills all three gaps within a single integrated empirical framework.

1.3 Research Objectives

The primary objective is to design, implement, and rigorously validate an automated quantitative portfolio system integrating the Fama-French three-factor model with K-Means clustering and SLSQP optimization. Specific objectives include:

(1) Automating the synchronized data pipeline combining equity price data (yfinance API) with Fama-French factor data (Kenneth French Data Library) via time-aligned inner join, covering 15 S&P 500 constituents across 2,515 trading days from January 2014 to December 2023.

(2) Validating the stationarity of all 15 excess return series using the Augmented Dickey-Fuller (ADF) test as a prerequisite for valid OLS regression.

(3) Extracting rolling five-dimensional risk feature vectors $[\alpha, \beta_{Mkt}, \beta_{SMB}, \beta_{HML}, \sigma_{\varepsilon}]$ for each security within each training window.

(4) Implementing K-Means clustering (K=3) with k-means++ initialization and Centroid Sorting to produce consistent, economically interpretable cluster labels across all 108 rebalancing windows.

(5) Applying SLSQP optimization with Ledoit-Wolf covariance shrinkage, hard per-asset weight caps (≤15%), and a cluster-level concentration constraint (≤40% for Cluster 2), and evaluating performance across nine dimensions against three benchmark strategies.

1.4 Research Hypotheses

This study formally tests two hypotheses at the 5% significance level using multiple statistical methods:

H_1 (Sharpe Ratio Superiority): The FF-KMeans-SLSQP strategy achieves a statistically significantly higher out-of-sample Sharpe ratio than each of the three benchmark strategies: pure SLSQP without clustering, 1/N equal weighting, and the SPY ETF.

H_2 (Distributional Differentiation): The daily return distribution of the FF-KMeans-SLSQP strategy differs significantly from that of the SPY ETF during periods of elevated market volatility, specifically the COVID-19 crash (Q1 2020), the post-COVID recovery (Q2 2020–Q4 2021), and the Federal Reserve tightening cycle (2022).

Chapter II: Literature Review

2.1 Capital Asset Pricing Model: Foundations and Empirical Limitations

The Capital Asset Pricing Model, developed independently by Sharpe (1964) and Lintner (1965) and given axiomatic foundations by Mossin (1966), established the equilibrium relationship between an asset's expected excess return and its beta with the market portfolio. The model posits that in an efficient market with homogeneous expectations and frictionless trading, all investors hold the tangency portfolio (the market portfolio), and expected returns are determined solely by market covariance risk. CAPM's elegance and parsimony positioned it as the dominant asset pricing framework in both academic research and practitioner applications throughout the 1970s and 1980s.

However, the accumulation of anomalous empirical evidence challenged CAPM's sufficiency as a complete description of cross-sectional return variation. Banz (1981) documented the size effect, demonstrating that smaller-capitalization stocks earn higher risk-adjusted returns than their CAPM-implied expected returns, even after controlling for market beta. Rosenberg et al. (1985) established that stocks with high book-to-market ratios (value stocks) earn significant excess returns relative to low book-to-market stocks (growth stocks), a finding that could not be attributed to higher market beta. Stattman (1980) and Fama and French (1992) confirmed and extended these findings across broad U.S. equity cross-sections, with Fama and French (1992) demonstrating that beta itself has minimal incremental explanatory power for returns once size and book-to-market ratio are included as controls. These systematic anomalies motivated the development of multifactor asset pricing models.

2.2 Fama-French Three-Factor Model

Fama and French (1993) formalized the size and value effects by constructing two factor-mimicking portfolios: SMB (Small Minus Big), which captures the return differential between small- and large-capitalization stocks, and HML (High Minus Low), which captures the premium between high and low book-to-market stocks. The model is estimated as a time-series regression for each security i:

$$r_{excess,i,t} = \alpha_i + \beta_{Mkt,i}(Mkt - RF_t) + \beta_{SMB,i}(SMB_t) + \beta_{HML,i}(HML_t) + \varepsilon_{i,t}$$

where $r_{excess,i,t} = r_{i,t} - RF_t$ is the daily excess return, (Mkt−RF)*t, SMB_t, and HML_t are the realized factor returns on day t, α_i is the OLS intercept (Jensen's alpha), and* $\varepsilon_{i,t}$ is the

zero-mean regression residual. This specification is the empirical form adopted in the present study's feature extraction stage (Section 3.4). Using 25 size-B/M-sorted portfolios as test assets, Fama and French (1993) demonstrated that the three-factor model explains 90–95% of time-series return variation, compared to 70–80% for CAPM alone. Subsequent extensions include the five-factor model (Fama & French, 2015), which adds profitability (RMW) and investment (CMA) factors, and the Carhart (1997) four-factor model, which incorporates momentum (WML) following Jegadeesh and Titman (1993). International evidence confirms value premia in 12 of 13 markets (Fama & French, 1998).

The economic interpretation of SMB and HML remains contested. Fama and French (1993, 1996) argue that both factors proxy for systematic risk: HML loads on financially distressed firms with elevated sensitivity to economic downturns, and the value premium represents rational risk compensation. In contrast, Daniel and Titman (1997) and Lakonishok, Shleifer, and Vishny (1994) attribute the premia to investor mispricing and the overextrapolation of past earnings growth rather than to priced risk exposures. This distinction has a direct implication for the present study: under the risk-based view, K-Means clusters defined by β_{SMB} and β_{HML} groupings identify portfolios with genuinely distinct systematic risk profiles; under the mispricing view, the same clusters reflect behavioral categories. Importantly, the empirical validity of the clustering structure is independent of which interpretation is adopted.

2.3 Modern Portfolio Theory and Mean-Variance Optimization

Markowitz (1952) established the formal mathematical framework for portfolio construction by characterizing the mean-variance efficient frontier, the set of portfolios maximizing expected return for each level of portfolio variance. The insight that diversification reduces idiosyncratic risk without sacrificing expected return provided the theoretical foundation for modern portfolio management. The maximum Sharpe ratio portfolio (Sharpe, 1966), defined as the portfolio maximizing the ratio of excess expected return to standard deviation, represents the optimal risky portfolio for investors facing a risk-free asset.

In practice, mean-variance optimization is well known to produce unstable, concentrated portfolios that underperform out of sample. Chopra and Ziemba (1993) quantified the relative costs of estimation errors, finding that errors in expected returns are approximately 11 times more costly than errors in variances in their impact on portfolio

welfare loss (for a medium-risk-aversion investor). Merton (1980) demonstrated theoretically that for typical equity return series with annualized volatility of 15–20% and daily observations, the standard error of the historical mean estimator is so large relative to any true return edge that the expected return vector is effectively noise-dominated. This observation has profound implications for SLSQP optimization: the optimizer's effective direction of travel is toward minimum-variance, even when nominally solving a maximum-Sharpe problem.

De Miguel et al. (2009) conducted an influential empirical investigation comparing 14 portfolio optimization models against the 1/N equal-weight benchmark across seven empirical datasets. Their striking finding, that no optimization strategy consistently outperforms 1/N out of sample over reasonable evaluation horizons, was attributed primarily to estimation error in both the expected return vector and the covariance matrix. This result underscores the importance of both robust covariance estimation and rigorous out-of-sample evaluation protocols, both of which are central to the present study's methodology.

2.4 Covariance Matrix Estimation: Ledoit-Wolf Shrinkage

The sample covariance matrix, while a consistent estimator in the asymptotic limit, suffers from severe finite-sample inefficiency when the ratio of assets to observations (n/T) is non-negligible. In such cases, extreme eigenvalues of the sample covariance matrix are systematically biased, where large eigenvalues are inflated, small eigenvalues are deflated, leading to numerical near-singularity and poor out-of-sample portfolio performance (Ledoit & Wolf, 2004).

Ledoit and Wolf (2004) proposed an analytically optimal linear shrinkage estimator that combines the sample covariance matrix with a structured target (the scaled identity matrix in their analytical shrinkage version) using a data-driven shrinkage coefficient that minimizes the expected Frobenius norm of the estimation error. The Ledoit-Wolf estimator has been shown to substantially improve covariance matrix conditioning and, consequently, the stability and out-of-sample performance of mean-variance optimized portfolios. In the present study, the training window of $T = 252$ trading days and universe of $n = 15$ assets yields a ratio of $T/n \approx 17$, a regime in which Ledoit-Wolf shrinkage provides meaningful regularization relative to the sample covariance estimate.

2.5 Machine Learning in Asset Pricing and Portfolio Construction

The intersection of machine learning and empirical finance has grown substantially in the past decade, driven by both increased data availability and advances in statistical learning methodology. Gu et al. (2020) provided the most comprehensive evaluation to date, applying a broad suite of ML models, including penalized linear models (LASSO, Ridge), gradient boosted trees, random forests, and deep neural networks, to monthly U.S. equity return prediction across approximately 30,000 equities from 1957 to 2016. Their results demonstrate consistent and statistically significant out-of-sample predictability from machine learning relative to traditional linear factor models, with neural networks achieving the highest R^2 values. Gu et al. (2020) also documented that the most important predictive features correspond closely to variables identified by the theoretical factor pricing literature, such as market capitalization, book-to-market ratio, and short-term reversal, validating the present study's strategy of anchoring the ML feature space in Fama-French factor exposures.

Within portfolio construction specifically, Lopez de Prado (2016) introduced Hierarchical Risk Parity (HRP), a clustering-based portfolio construction method that uses hierarchical clustering of the correlation matrix to define diversification structure without requiring matrix inversion. HRP is particularly robust to covariance matrix estimation error and has attracted significant practitioner attention as an alternative to classical optimization. The present study employs K-Means clustering operating on Fama-French factor coefficients rather than on correlations, representing a conceptually distinct approach that leverages economic theory to motivate the feature space for clustering rather than relying solely on return covariation patterns. A direct empirical comparison between these two clustering paradigms on the same dataset constitutes a natural direction for future research.

2.6 K-Means Clustering: Algorithm and Applications in Finance

K-Means, formalized by MacQueen (1967), is an iterative partitional clustering algorithm that minimizes the sum of squared Euclidean distances from each observation to its assigned cluster centroid (within-cluster inertia). The algorithm alternates between an assignment step, assigning each point to the nearest centroid, and an update step, recomputing centroids as cluster means, until convergence to a local optimum. The initialization strategy k-means++ (Arthur & Vassilvitskii, 2007) addresses the classical K-Means sensitivity to initial centroid placement by selecting initial centroids with probability proportional to squared distance from previously placed centroids, substantially reducing convergence to poor local optima at moderate computational overhead.

Applications of K-Means to financial data have examined clustering on various feature spaces. Clustering on fundamental financial ratios has been shown to recover known sector and style classifications (See Roncalli, 2013, Ch 7 for a review). Applications to risk factor exposures as features are relatively less common in the published literature, making the present study's approach, using Fama-French OLS regression coefficients as the clustering feature space, a methodological contribution. The economic interpretability of clusters defined by factor loadings (e.g., high-HML = Value cluster, low-HML = Growth cluster) is a significant advantage over clustering on raw prices or correlations, enabling ex-post validation of cluster structure against known economic categories.

2.7 Walk-Forward Validation in Quantitative Finance

Walk-Forward Validation (WFV), also termed rolling-window cross-validation or time-series cross-validation, is the methodologically appropriate evaluation framework for adaptive quantitative strategies applied to financial time series. Unlike static train-test split designs, which implicitly use all future data to construct the test set boundary and k-fold cross-validation designs that violate temporal ordering by using future observations to evaluate past predictions, WFV strictly respects the causal structure of financial data: at any point in time, only historically observed information is used in model fitting (Lopez de Prado, 2018).

Lopez de Prado (2018) formalized several necessary conditions for valid financial backtesting, including the requirements that all data transformations (normalization, factor estimation) be performed exclusively within each training window, and that no parameter selection be informed by out-of-sample performance. Bailey and Lopez de Prado (2012) documented the problem of backtest overfitting, the inflation of backtested Sharpe ratios through implicit or explicit multiple comparisons of strategy variants and derived the minimum sample size required to achieve given levels of statistical confidence in the reported performance. The present study strictly adheres to WFV principles by re-fitting StandardScaler, OLS regression, and K-Means clustering within each of the 108 training windows, and by fixing all hyperparameters (train window = 252 days, rebalance period = 21 days, K = 3) prior to the backtest run.

2.8 Statistical Testing of Sharpe Ratio Differences

Lo (2002) derived the asymptotic distribution of the Sharpe ratio estimator under the general assumption of stationary, ergodic return distributions with potentially non-zero

skewness and excess kurtosis. His key result demonstrates that the standard error of the annualized Sharpe ratio depends not only on sample size T but also on higher-order moments of the return distribution, and that the standard error of the Sharpe ratio difference between two correlated strategies depends on the correlation between return series. Lo (2002) developed an asymptotic z-test for Sharpe ratio differences that is better calibrated for financial return data than the standard t-test on daily return differentials, which implicitly assumes independent and identically distributed normal returns.

The practical importance of Lo's (2002) framework is illustrated by the present study's power analysis: detecting a Sharpe differential of $\Delta SR = 0.048$ (the estimated increment from K-Means clustering) at 80% power in a daily return series requires approximately $T = 8{,}364$ trading days ≈ 33.2 years of data — a fundamentally infeasible requirement for any empirical financial study. This finding generalizes the observation of Bailey and Lopez de Prado (2012) that the minimum required observations for statistical significance of Sharpe ratio comparisons scales as $O(1/\Delta SR^2)$, making small Sharpe increments statistically invisible even in large samples. Appropriate economic interpretation of small Sharpe differentials therefore requires explicit power analysis, as conducted in the present study.

Chapter III: Methodology

3.1 System Architecture

The proposed system is structured as a five-stage closed-loop pipeline embedded within a Walk-Forward Validation framework. Stages 1 through 3 constitute the In-Sample Learning phase, in which all model components are fitted exclusively on historical training data; Stages 4 and 5 constitute the Out-of-Sample Application phase, in which fitted models generate portfolio weights for the subsequent rebalancing period. The pipeline is executed 108 times sequentially over the backtest horizon, with no information from any stage crossing the training-test boundary.

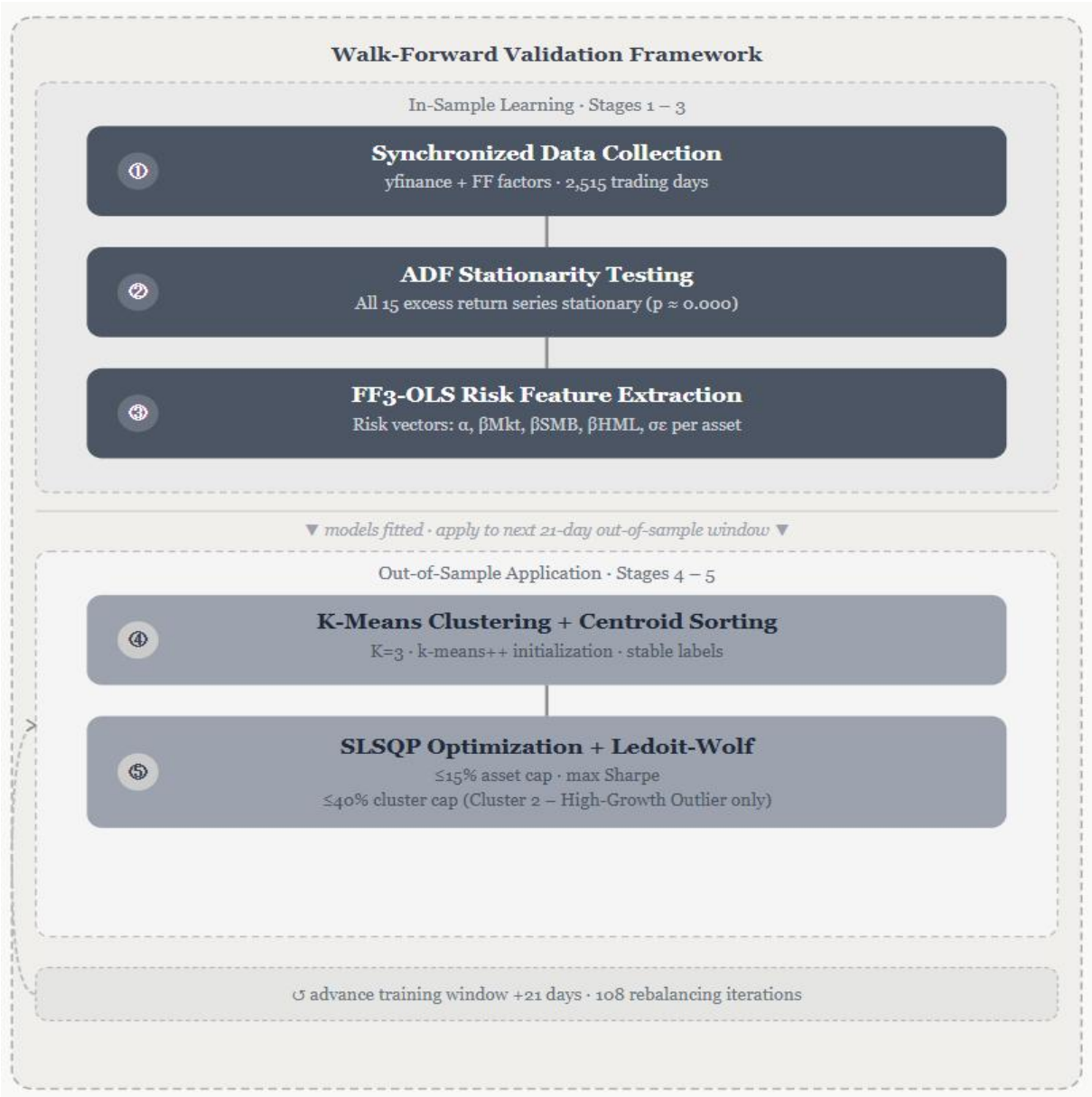


Figure 1: Five-Stage Pipeline Architecture.

Stages: (1) Synchronized Data Collection → (2) ADF Stationarity Testing → (3) FF3-OLS Feature Extraction → (4) K-Means Clustering with Centroid Sorting → (5) SLSQP Optimization with Ledoit-Wolf Shrinkage + Hard Constraints. Walk-Forward loop advances by 21 trading days per iteration.

3.2 Data Collection and Preprocessing

3.2.1 Asset Universe and Selection Criteria

The study universe comprises 15 common equities selected from the S&P 500 index according to four quantitative criteria: (1) continuous S&P 500 index membership from January 1, 2014 through December 31, 2023, ensuring survivorship consistency within the study period; (2) average daily trading volume exceeding 5 million shares across the full study period, ensuring practical execution feasibility at institutional AUM levels; (3) representation across a minimum of six GICS sectors, ensuring multidimensional sectoral diversification; and (4) complete daily adjusted closing price availability on the yfinance API with no data gaps exceeding five consecutive trading days.

Ticker	Company	GICS Sector	Rationale for Inclusion
AAPL	Apple Inc.	Information Technology	Largest U.S. market cap; growth/negative HML benchmark
MSFT	Microsoft Corporation	Information Technology	Enterprise software diversifier; stable large-cap growth
NVDA	NVIDIA Corporation	Information Technology	GPU/AI outlier; extreme β_Mkt and β_HML exposure
GOOGL	Alphabet Inc.	Communication Services	Advertising-driven megacap; low beta relative to peer tech
JPM	JPMorgan Chase & Co.	Financials	Largest U.S. bank by assets;

Ticker	Company	GICS Sector	Rationale for Inclusion
			positive HML (value)
GS	Goldman Sachs Group	Financials	Investment banking/trading; cyclical value exposure
BAC	Bank of America Corp.	Financials	Retail banking; high HML and SMB loading
JNJ	Johnson & Johnson	Health Care	Defensive healthcare; lowest market beta in universe (0.59)
UNH	UnitedHealth Group	Health Care	Managed care; strong alpha and low volatility
AMZN	Amazon.com Inc.	Consumer Discretionary	E-commerce/cloud; negative HML (hyper-growth)
HD	Home Depot Inc.	Consumer Discretionary	Retail; near-zero HML, classified in Value cluster
XOM	Exxon Mobil Corporation	Energy	Commodity-linked; positive HML value exposure
CVX	Chevron Corporation	Energy	Energy major; mirrors XOM factor profile
CAT	Caterpillar Inc.	Industrials	Capital goods cyclical; positive SMB and HML
META	Meta Platforms Inc.	Communication Services	Social media; negative HML, high idiosyncratic risk

Table 1: Stock Universe by Sector (15 S&P 500 Constituents, 2014–2023)

Note. All 15 securities maintained continuous S&P 500 membership from January 2014 through December 2023 and satisfied minimum average daily trading volume of 5 million shares. Data sourced via yfinance API.

3.2.2 Return Computation and Synchronization

Daily logarithmic returns are computed for each security as $r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right)$, where $P_{i,t}$ is the adjusted closing price on day t (incorporating splits and dividends via the auto_adjust=True parameter). Logarithmic returns are preferred over simple returns for their time-additivity property and their closer approximation to normality in finite samples. Excess returns are defined as $r_{excess,i,t} = r_{i,t} - RF_t$, where RF_t is the daily risk-free rate sourced from the Kenneth French Data Library. Time alignment between equity price data and Fama-French factor data is implemented via an inner join on the date index, yielding 2,515 trading days afteralignment (January 3, 2014 – December 29, 2023).

3.3 Stationarity Testing: Augmented Dickey-Fuller Test

Prior to OLS regression, stationarity of all 15 excess return series is verified using the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979). The null hypothesis H_0 posits that the series contains a unit root (i.e., is non-stationary); rejection at the 5% significance level ($p < 0.05$) is required to confirm stationarity. The ADF test regresses the first difference of the series on lagged levels and lagged first differences, with lag order selected by the Akaike Information Criterion. Stationarity of excess return series is theoretically expected, as logarithmic returns represent the first difference of log prices, which are commonly modeled as integrated of order one (I(1)).

3.4 Risk Feature Extraction via Fama-French OLS Regression

Within each training window of 252 trading days, the following time-series regression is estimated separately for each security i using Ordinary Least Squares:

$$r_{excess,i,t} = \alpha_i + \beta_{Mkt,i}(Mkt - RF_t) + \beta_{SMB,i}(SMB_t) + \beta_{HML,i}(HML_t) + \varepsilon_{i,t}$$

OLS estimation yields the five-dimensional risk feature vector for each security: $f_i = [\alpha_i,\ \beta_{Mkt,i},\ \beta_{SMB,i},\ \beta_{HML,i},\ \sigma_{\varepsilon,i}]$, where $\sigma_{\varepsilon,i}$ denotes the standard deviation of OLS residuals, representing the idiosyncratic (firm-specific) risk component unexplained by the three systematic factors. Each regression uses only the 252 observations within the current training window, strictly enforcing the no-data-leakage constraint. The feature vector captures five

distinct dimensions of security risk: abnormal return (alpha), systematic market exposure, size tilt, value/growth orientation, and idiosyncratic volatility.

3.5 K-Means Clustering with Centroid Sorting

The 15-by-5 feature matrix $F = \{\mathrm{f}_i\}_{i=1}^{15}$ is standardized using StandardScaler (zero mean, unit variance) fitted exclusively on training data before K-Means is applied. Standardization is essential to prevent features with larger absolute scales (e.g., $\beta_{Mkt} \approx 1.0$) from dominating features with smaller scales (e.g., $\alpha \approx 0.001$) in the Euclidean distance computation.

K-Means minimizes the within-cluster inertia objective: $J = \sum_k \sum_{\mathbf{x} \in C_k} \|\mathrm{x} - \mu_k\|^2$, where C_k is cluster k and μ_k is its centroid. The algorithm is initialized using k-means++ (Arthur & Vassilvitskii, 2007) with n_init = 15 independent random restarts to minimize local-optima sensitivity. A global random seed (RANDOM_STATE = 42) ensures complete reproducibility across all rebalancing windows.

K = 3 clusters is fixed by design, providing three economically interpretable risk profiles. Although the silhouette coefficient analysis favors K = 2, this configuration produces a degenerate 14:1 asset split driven entirely by NVDA's extreme factor exposure (β_{Mkt} = 1.61), rendering it economically uninformative. K = 3 creates three meaningful risk groups while maintaining K = 4's silhouette score within the margin of statistical noise (0.339 vs. 0.343). Centroid Sorting re-labels cluster indices by ascending mean β_{Mkt} after each K-Means fit, ensuring deterministic and economically consistent cluster labels across all 108 rebalancing windows.

3.6 SLSQP Portfolio Optimization

Portfolio weights are determined by maximizing the expected Sharpe ratio subject to institutional constraints. The optimization problem is formulated as:

$$\min_{\mathrm{w}} \quad -SR(\mathrm{w}) = -\frac{\mathrm{w}^\top \mu \cdot \sqrt{252}}{\sqrt{\mathrm{w}^\top \Sigma_{LW}\, \mathrm{w}} \cdot \sqrt{252}}$$

subject to: $\sum_i w_i = 1$ (full investment), $0 \le w_i \le 0.15 \quad \forall\, i$ (per-asset cap), $\sum_{i \in C_2} w_i \le 0.40$ (cluster cap), The per-asset cap of 15% enforces minimum diversification across at least seven securities, equivalent to 2.25× the equal-weight allocation of 6.67%. The cluster cap of 40% is applied to Cluster 2 (NVDA singleton) to prevent extreme single-security concentration

during high-alpha periods; this value follows standard institutional practice for single-sector exposure limits, though sensitivity to this specific threshold was not formally analyzed.

The expected return vector μ is estimated as the historical mean of training-window daily excess returns. As Merton (1980) emphasized, this estimator is noise-dominated, implying that the optimization primarily targets minimum-variance. The covariance matrix Σ is estimated using the Ledoit-Wolf analytical shrinkage estimator (Ledoit & Wolf, 2004), combining the sample covariance matrix with the scaled identity matrix using an analytically derived optimal shrinkage coefficient. SLSQP (Sequential Least Squares Programming) is applied via scipy.optimize.minimize with a maximum of 1,500 iterations and convergence tolerance ftol = 1×10^{-10}. A Fallback to Inverse Volatility Weighting is activated in the event of SLSQP non-convergence.

3.7 Walk-Forward Validation Framework

The Walk-Forward Validation framework is parameterized as follows: training window T = 252 trading days; rebalancing period = 21 trading days (approximately one calendar month); total rebalancing windows = 108, covering the out-of-sample period from January 5, 2015 through December 29, 2023 (2,263 trading days). Transaction costs of 10 basis points per rebalancing period are deducted as a flat cost on the first trading day of each period, applied to all three active strategies. A parallel proportional TC calculation $TC_{prop} = 10\,bps \times \sum_i \left|w_i^{new} - w_i^{old}\right|$ is computed for verification purposes. At each window, the complete pipeline, StandardScaler, OLS regression, K-Means clustering, and SLSQP optimization, is re-executed from scratch on training data only.

3.8 Performance Metrics

Out-of-sample performance is evaluated across nine quantitative metrics, covering return, risk, risk-adjusted return, drawdown, recovery, and active management dimensions:

Metric	Formula	Interpretation
Total Return	$TR = \frac{NAV_T}{NAV_0} - 1$	Cumulative dollar gain over full backtest period
Geometric Annual Return	$R_{ann} = (1 + TR)^{1/T_{years}} - 1$	CAGR; accounts for compounding effects

Metric	Formula	Interpretation
Annual Volatility	$\sigma_{ann} = \sigma_{daily} \times \sqrt{252}$	Realized annualized standard deviation of daily returns
Sharpe Ratio	$SR = \frac{R_{ann} - R_f}{\sigma_{ann}}$	Risk-adjusted return per unit total risk
Sortino Ratio	$Sortino = \frac{R_{ann} - R_f}{\sigma_{downside} \times \sqrt{252}}$	Penalizes only downside semi-variance, not upside volatility
Calmar Ratio	$Calmar = \frac{R_{ann}}{\lvert MDD \rvert}$	Return per unit of maximum peak-to-trough loss
Maximum Drawdown	$MDD = \min_t \left(\frac{NAV_t}{\max_{s \le t} NAV_s} - 1 \right)$	Worst peak-to-trough decline over the backtest period
Max Recovery (Completed)	$MR_{completed} = \max_{episodes} (t_{recovery} - t_{peak})$ across all fully recovered episodes, in trading days	Longest observed recovery from drawdown trough back to prior equity peak; reported in Figure 6
Max Recovery (Outstanding)*	$MR_{outstanding} = T_{end} - t_{peak}$	Trading days elapsed since latest all-time high for strategies still in drawdown at December 29, 2023; reported in Table 7 with asterisk
Information Ratio (vs SPY)	$IR = \frac{R_{active}}{\sigma_{TE}}$	Active management quality metric; IR ≥ 0.5 is "good" (Grinold & Kahn, 2000)

Table 2: Performance Metrics Definitions

*Applies only to strategies that had not recovered to their latest equity peak by end-of-sample. This metric understates true recovery duration as the drawdown episode remained open.

3.9 Hyperparameter Sensitivity Analysis

Robustness of results to hyperparameter specification is assessed through a 3×3 grid search over training window length (126, 252, 504 trading days) and rebalancing period (10, 21, 63 trading days), generating nine independent Walk-Forward backtest runs. The baseline configuration (Train=252, Rebal=21) is designated a priori based on theoretical conventions established in the factor model estimation literature (Grinold & Kahn, 2000), and is not selected by optimization over the sensitivity grid. Reporting the full 9-cell grid mitigates the risk of inadvertent data-snooping in configuration selection.

3.10 Statistical Hypothesis Testing Framework

H_1 (Sharpe ratio superiority) is tested using a sequential four-method protocol. First, the Durbin-Watson (DW) statistic (Durbin & Watson, 1950) is computed on the daily return differential series to verify the absence of serial autocorrelation, which would invalidate standard t-tests. Second, a standard one-sided t-test is applied to the differential return series. Third, a Newey-West HAC-adjusted t-test (Newey & West, 1987) with Bartlett kernel and 21 lags (corresponding to one rebalancing period) is applied to produce heteroskedasticity- and autocorrelation-consistent standard errors. Fourth, the Lo (2002) asymptotic z-test is applied directly to the annualized Sharpe ratio estimates. Bootstrap confidence intervals (B = 3,000 resamples with replacement) are computed for each strategy's annualized Sharpe ratio. Statistical power for each comparison is computed using the non-central t-distribution with Cohen's d (Cohen, 1988) computed from the differential return series.

H_2 (return distribution differentiation) is tested using the two-sample Kolmogorov-Smirnov (KS) test applied to daily return distributions during three market regime sub-periods: the COVID-19 Crash (January 20 – March 23, 2020), the post-COVID Recovery (Q2 2020 – Q4 2021: April 1, 2020 – December 31, 2021), and the Federal Reserve Tightening Cycle (2022: January 3 – December 30, 2022). Supplemental tail risk metrics, Value at Risk (VaR) at the 95th percentile and Conditional Value at Risk (CVaR, Expected Shortfall) at the 95th percentile — are reported for each sub-period and each strategy.

Chapter IV: Results

4.1 ADF Stationarity Test Results

ADF tests applied to all 15 daily excess return series universally reject the null hypothesis of non-stationarity at the 1% significance level, with p-values of approximately 0.000 across all securities (Table 3). ADF statistics range from −50.915 (AMZN, reflecting the high-frequency variance structure of this series) to −9.952 (JPM). These results are consistent with the theoretical expectation that logarithmic daily returns, as first differences of log prices, follow stationary I(0) processes. The universal rejection of the unit root null hypothesis confirms the validity of the OLS regression framework applied in Stage 3 of the pipeline.

Ticker	Company	ADF Stat	p-value	Lag	Result
AAPL	Apple Inc.	−15.822	0.000	8	✓ Stationary
MSFT	Microsoft Corp.	−17.513	0.000	8	✓ Stationary
NVDA	NVIDIA Corp.	−17.964	0.000	7	✓ Stationary
GOOGL	Alphabet Inc.	−11.469	0.000	21	✓ Stationary
JPM	JPMorgan Chase	−9.952	0.000	27	✓ Stationary
GS	Goldman Sachs	−16.964	0.000	7	✓ Stationary
BAC	Bank of America	−15.681	0.000	8	✓ Stationary
JNJ	Johnson & Johnson	−16.335	0.000	9	✓ Stationary
UNH	UnitedHealth Group	−15.437	0.000	13	✓ Stationary
AMZN	Amazon.com Inc.	−50.915	0.000	0	✓ Stationary
HD	Home Depot Inc.	−17.205	0.000	8	✓ Stationary
XOM	Exxon Mobil Corp.	−17.809	0.000	7	✓ Stationary

Ticker	Company	ADF Stat	p-value	Lag	Result
CVX	Chevron Corp.	−17.490	0.000	7	✓ Stationary
CAT	Caterpillar Inc.	−15.110	0.000	10	✓ Stationary
META	Meta Platforms	−17.334	0.000	9	✓ Stationary

Table 3: ADF Stationarity Test Results for 15 Excess Return Series

Note. ADF = Augmented Dickey-Fuller. H_0: series contains a unit root. All p-values computed by MacKinnon (1994) response surface approximation. Lag = number of augmented lags selected by AIC. 15/15 series reject H_0 at $\alpha = 0.01$.

4.2 Fama-French Risk Feature Matrix

Table 4 presents the five-dimensional Fama-French OLS regression coefficients estimated over the full 10-year dataset (2014–2023, n = 2,515 observations). These full-period estimates serve as a descriptive illustration; the rolling-window estimates used in each of the 108 Walk-Forward periods differ across rebalancing windows as market regimes evolve.

Ticker	α (daily)	β_{Mkt}	β_{SMB}	β_{HML}	ε_i
AAPL	0.000321	1.1744	−0.3287	−0.4087	0.0113
MSFT	0.000329	1.1993	−0.4654	−0.4700	0.0093
NVDA	0.001092	1.6148	+0.1301	−0.8637	0.0204
GOOGL	0.000028	1.1299	−0.2477	−0.4263	0.0113
JPM	0.000019	1.1464	−0.1260	+0.8733	0.0084
GS	−0.000152	1.1901	+0.0669	+0.7312	0.0098
BAC	−0.000181	1.2691	+0.0484	+1.0465	0.0099

Ticker	α (daily)	β_{Mkt}	β_{SMB}	β_{HML}	ε_i
JNJ	−0.000014	0.5921	−0.4267	+0.0434	0.0091
UNH	0.000364	0.8964	−0.3945	+0.0921	0.0125
AMZN	0.000176	1.1231	−0.1813	−0.7629	0.0148
HD	0.000168	0.9723	−0.0178	−0.0008	0.0105
XOM	−0.000227	0.9105	−0.0282	+0.9096	0.0118
CVX	−0.000220	1.0438	−0.0521	+0.9299	0.0124
CAT	0.000118	1.0413	+0.2000	+0.6811	0.0128
META	0.000065	1.2448	−0.0847	−0.6743	0.0180

Table 4: Fama-French Three-Factor Risk Feature Matrix (Full Period, 2014–2023)

Note. α = daily Jensen's alpha (intercept). β_{Mkt}, β_{SMB}, β_{HML} = Fama-French factor loadings estimated by OLS. σ_ε = standard deviation of OLS residuals (idiosyncratic volatility). Fama-French factor data sourced from Kenneth French Data Library (Fama & French, 2024). Full-period estimation for descriptive purposes; rolling-window estimates differ across rebalancing periods.

4.3 K-Means Clustering Results

4.3.1 Optimal K Selection

Table 5 presents the elbow plot inertia and silhouette coefficient values for K = 2 through K = 6, evaluated on the full-period standardized feature matrix. The silhouette analysis nominally favors K = 2 (score = 0.493), but this configuration produces a degenerate 14:1 asset split — 14 securities in one cluster and NVDA as a singleton — driven entirely by NVDA's extreme β_{Mkt} exposure (1.615) relative to the universe mean (1.12). This degenerate

split provides no economically useful diversification structure. K = 3 (silhouette = 0.339) creates three economically interpretable risk profiles while K = 4 (silhouette = 0.343) offers marginally higher silhouette at the cost of one additional cluster that cannot be given a clear economic interpretation. K = 3 is therefore fixed by design.

K	Within-Cluster Inertia	Silhouette Score	Assessment
2	48.1228	0.4932	Highest silhouette; degenerate 14:1 split (NVDA outlier)
3 ★	26.7679	0.3386	SELECTED — 3 economically meaningful risk clusters
4	18.3682	0.3429	Marginally higher silhouette; fourth cluster lacks clear interpretation
5	12.7521	0.3353	Declining silhouette; over-partitioning of small universe
6	9.5771	0.2576	Sharp silhouette decline; clusters not statistically separable

Table 5: Elbow Plot Inertia and Silhouette Score by Number of Clusters (K = 2–6)

Note. ★ = selected configuration. Inertia = total within-cluster sum of squared Euclidean distances to centroids, computed on StandardScaler-normalized feature matrix $[\alpha, \beta_{Mkt}, \beta_{SMB}, \beta_{HML}, \sigma_{\varepsilon}]$. Silhouette score ranges from −1 (poor) to +1 (perfect cluster separation). K = 3 enforced over K = 2 on economic grounds despite lower silhouette.

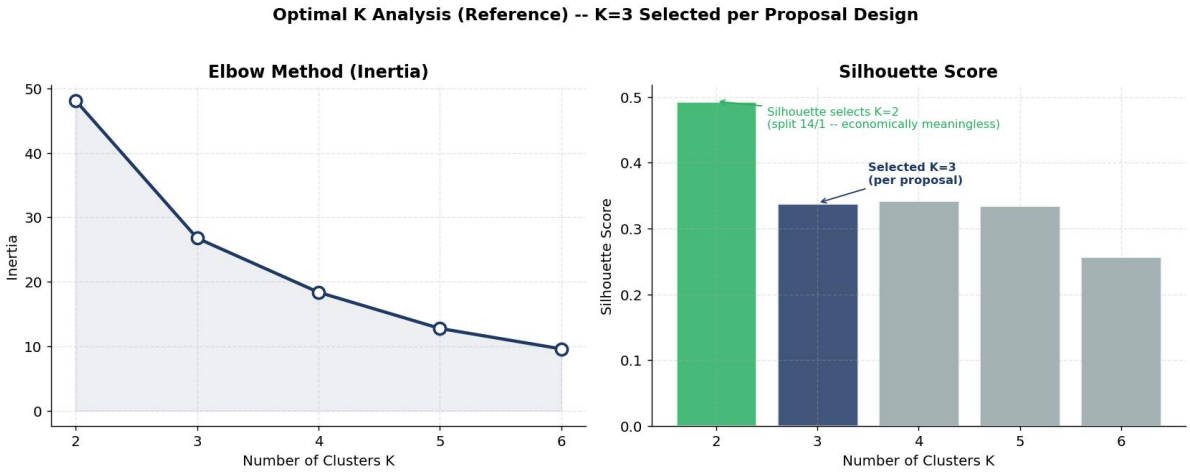


Figure 2: Elbow Plot and Silhouette Analysis (K = 2–6).

Left: within-cluster inertia (rapid decline K=2→3, then diminishing returns). Right: silhouette score by K (K=2 nominally optimal at 0.493 but yields degenerate 14:1 split; K=3 selected, score = 0.339).

4.3.2 Cluster Composition and Economic Interpretation

Table 6 presents the composition and centroid characteristics of the three K-Means clusters formed on the full-period standardized feature matrix. The principal axis of cluster separation is β_{HML}, not β_{Mkt}: Cluster 0 (Growth) contains securities with negative average HML loadings ($\beta_{HML} = -0.372$), while Cluster 1 (Value) contains securities with strongly positive average HML loadings ($\beta_{HML} = +0.739$). This emergent separation corresponds precisely to the Growth/Value dichotomy that is the defining theoretical contribution of the Fama-French HML factor, demonstrating that K-Means independently recovers this economic classification from the factor regression data without any explicit supervision or labeling.

Cluster	Economic Label	β_{Mkt}	β_{HML}	Constituent Securities
0 (Growth)	*Growth (Negative HML)*	1.051	−0.372	AAPL, MSFT, GOOGL, JNJ, UNH, AMZN, META (7 assets)
1 (Value)	*Value (Positive HML)*	1.082	+0.739	JPM, GS, BAC, HD, XOM, CVX, CAT (7 assets)
2 (Outlier)	*High-Growth Outlier*	1.615	−0.864	NVDA only (1 asset, Hard Constraint: ≤40% portfolio weight)

Table 6: K-Means Cluster Composition and Risk Centroid Characteristics (K = 3, Full Period)

Note. Cluster labels (Growth, Value, High-Growth Outlier) are assigned based on mean β_{HML} at the centroid. Centroid Sorting ensures Cluster 0 always has lowest mean β_{Mkt} and Cluster 2 has highest mean β_{Mkt}, guaranteeing label consistency across 108 rebalancing windows. Full-period estimates shown; rolling estimates vary across periods.

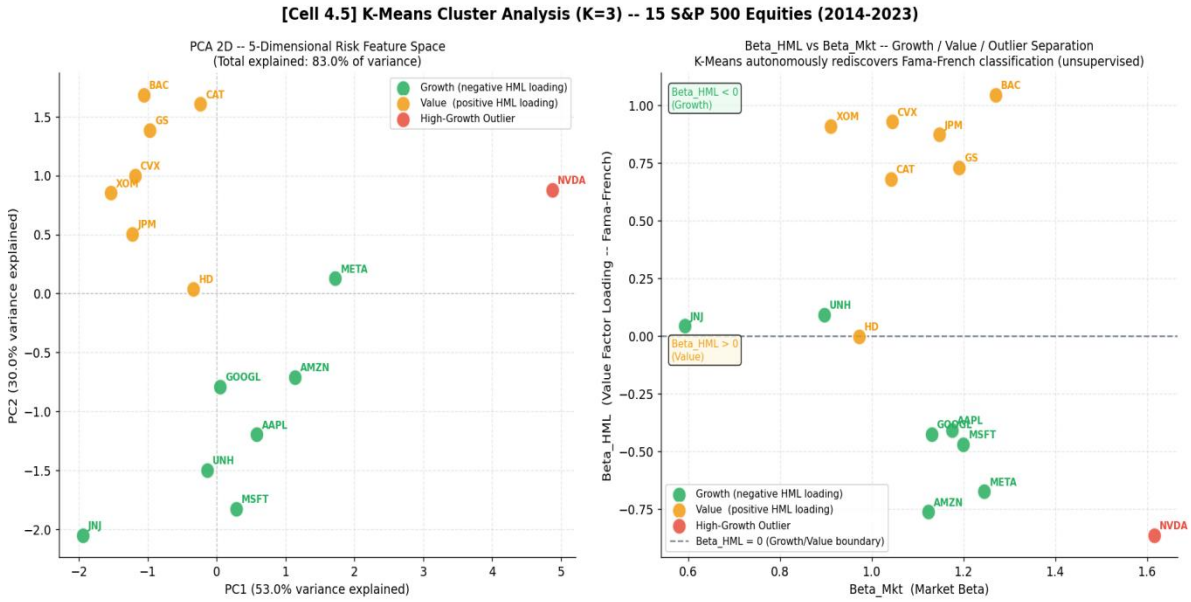


Figure 3: K-Means Cluster Visualization in PCA 2D Space (K = 3).

Points = 15 securities projected onto the first two PCs of the standardized $[\alpha, \beta_{Mkt}, \beta_{SMB}, \beta_{HML}, \sigma_{\varepsilon}]$ *feature space. Colors: blue = Growth (Cluster 0), orange = Value (Cluster 1), green = High-Growth Outlier (Cluster 2, NVDA). Primary separation axis in PCA space:* β_{HML}.

4.4 Overall Out-of-Sample Performance

Table 7 presents the nine-metric performance comparison across all four strategies over the full out-of-sample period (2015–2023, 2,263 trading days, 108 rebalancing windows). Transaction costs of 10 bps per rebalancing period are applied to all three active strategies (FF-KMeans-SLSQP, SLSQP Pure, and 1/N). SPY ETF is evaluated as a buy-and-hold passive strategy with no transaction costs applied.

Metric	① FF-KMeans-SLSQP	② SLSQP Pure	③ 1/N EW	④ SPY ETF
Total Return	234.35%	210.21%	216.44%	134.23%
Geometric Annual Return	14.39%	13.44%	13.69%	9.94%
Annual Volatility	21.17%	21.27%	21.13%	18.16%
Sharpe Ratio	0.6795	0.6316	0.6477	0.5475
Sortino Ratio	0.8238	0.7664	0.7789	0.6477
Calmar Ratio	0.4169	0.3894	0.3563	0.2781
Maximum Drawdown	−34.51%	−34.51%	−38.41%	−35.75%
Max Recovery (completed)	460d (1.8yr)	460d (1.8yr)	303d (1.2yr)	263d (1.0yr)
Days since latest peak*	~106d	~106d	~500d	~501d
Information Ratio vs SPY	0.4551	0.3540	0.6176	0.0000

Table 7: Out-of-Sample Performance Metrics: Four Strategies (January 2015 – December 2023)

Note. ① = FF-KMeans-SLSQP (primary model). ② = SLSQP optimization without K-Means clustering. ③ = 1/N equal-weight portfolio, rebalanced monthly. ④ = SPY ETF (SPDR S&P 500 ETF Trust, passive buy-and-hold benchmark). All active strategies include 10 bps flat transaction cost per rebalancing period. Out-of-sample period: January 5, 2015 – December 29, 2023. Max Recovery = trading days from drawdown peak to end-of-sample for unrecovered positions. * = position not yet recovered at end of sample: ① 106 days outstanding, ② 106 days, ③ 500 days, ④ 501 days.

*All four strategies had not recovered to their latest equity peak as of December 29, 2023 (end-of-sample). "Days since latest peak" measures trading days from the most recent all-time high to end-of-sample, distinct from the maximum completed recovery period reported above.

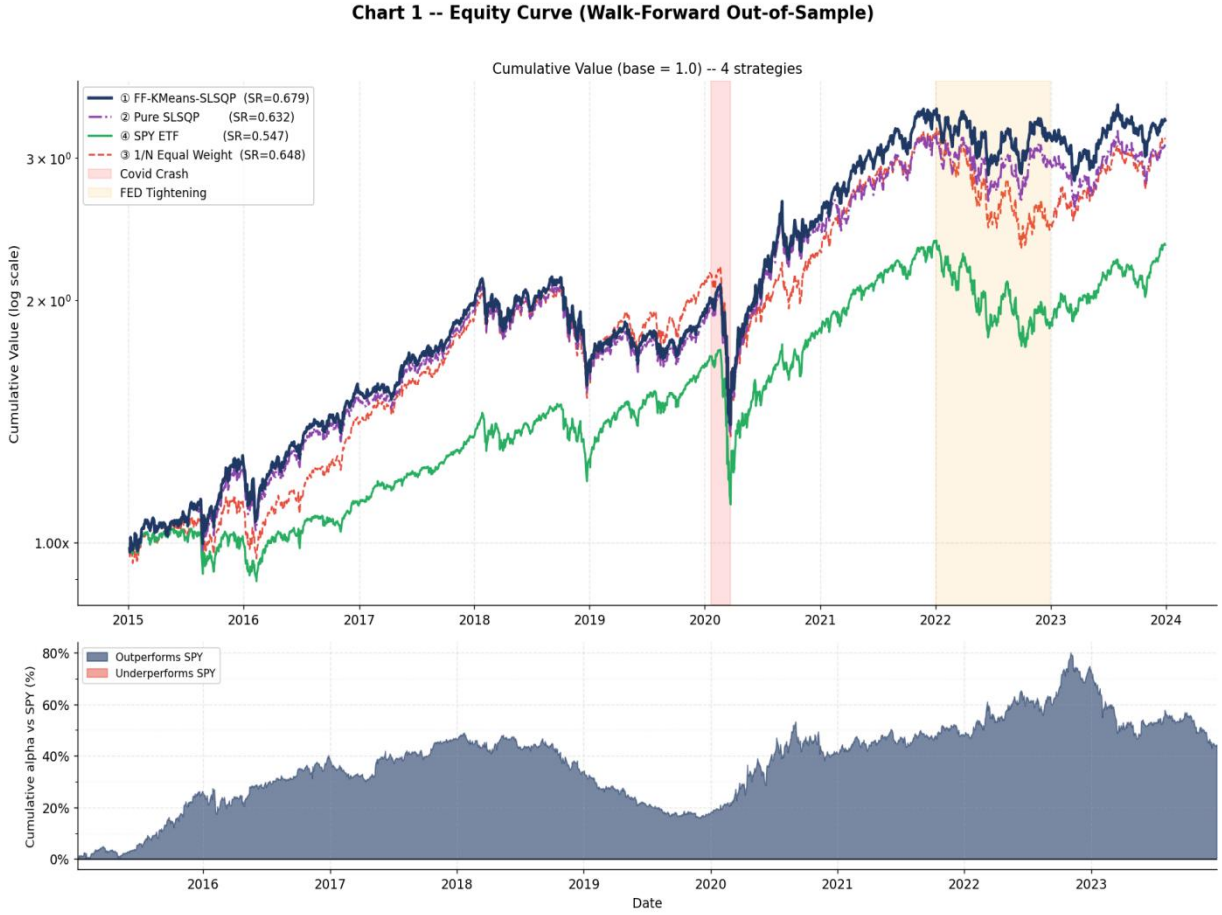


Figure 4: Equity Curve — Walk-Forward Out-of-Sample Performance (2015–2023).

Upper: log-scale cumulative wealth, normalized to 1.00 at Jan 5, 2015. Lower: rolling 21-day return differential (FF-KMeans-SLSQP vs. SPY ETF). Shaded: COVID-19 Crash (Q1 2020), Recovery (Q2 2020–Q4 2021), Fed Tightening (2022).



Figure 5: Rolling 252-Day Sharpe Ratio — Four Strategies (2015–2023).

Y-axis: [−2.0, 5.5] (data range: [−0.98, 4.74]). Strategy ① (FF-KMeans-SLSQP) maintains positive Sharpe ratios outside crisis periods; rolling windows shorter than 252 days shown as empty.

4.5 Incremental Contribution of K-Means Clustering

Isolating the contribution of K-Means clustering by comparing ① (FF-KMeans-SLSQP) against ② (pure SLSQP without clustering) reveals that the clustering step contributes positively across multiple dimensions (Table 8). The Sharpe ratio differential is ΔSharpe = +0.0478 (+7.6% relative improvement), reflecting the value of cluster-based diversification and the Hard Constraint in directing weights away from the single outlier security (NVDA). The Calmar ratio improves by +0.0275, and the Information Ratio versus SPY improves by +0.1011.

Metric	**① FF-KMeans**	**② SLSQP Pure**	**Δ (① − ②) and Interpretation**
Sharpe Ratio	0.6795	0.6316	+0.0478 (+7.6%)
Annual Return	14.39%	13.44%	+ 0.95 pp; clustering adds systematic return uplift
Calmar Ratio	0.4169	0.3894	+ 0.0275; better return per unit of max drawdown

Metric	① FF-KMeans	② SLSQP Pure	Δ (① − ②) and Interpretation
Information Ratio	0.4551	0.3540	+ 0.1011; stronger active management quality
Maximum Drawdown	−34.51%	−34.51%	≈ 0.00 pp; clustering does not alter max drawdown
Daily Return Correlation (r)	—	—	r = 0.9961 (R^2 = 0.992); independent variance = 0.78%

Table 8: Incremental Contribution of K-Means Clustering (Strategy ① vs. Strategy ②)

Note. Δ = Strategy ① minus Strategy ②. pp = percentage points. The near-unity correlation (r = 0.9961) confirms that K-Means value creation manifests in systematic risk-adjusted performance metrics rather than in daily return divergence. Both strategies achieve 100% SLSQP convergence (108/108 periods).

4.6 Portfolio Weight Allocation

Table 9 presents the average portfolio weight allocations across all 108 rebalancing periods for the FF-KMeans-SLSQP strategy. Defensive healthcare securities UNH (10.94%) and JNJ (10.09%) receive the highest average allocations, reflecting their combination of relatively low market beta (β_{Mkt} = 0.896 and 0.592, respectively) and positive alpha histories — consistent with the minimum-variance orientation of the Ledoit-Wolf-shrinkage-driven optimizer. NVDA receives a 9.41% average allocation, constrained at or near the 40% Cluster 2 cap in high-growth periods. High-volatility financial stocks BAC (1.93%) and GS (3.41%) receive minimal allocations, consistent with the optimizer's systematic down-weighting of high-idiosyncratic-risk securities.

Ticker	Cluster Assignment	Average Weight (%)	Std Dev (%)	Economic Rationale for Allocation Level
UNH	Growth (Cluster 0)	10.94	6.07	Highest alpha (0.000364/day); low β_{Mkt} (0.896)

Ticker	Cluster Assignment	Average Weight (%)	Std Dev (%)	Economic Rationale for Allocation Level
JNJ	Growth (Cluster 0)	10.09	6.52	Lowest β_{Mkt} in universe (0.592); defensive healthcare
NVDA	Outlier (Cluster 2)	9.41	6.39	Highest alpha (0.001092/day); capped at 40% cluster constraint
HD	Value (Cluster 1)	8.43	6.91	Near-zero β_{HML}; stable positive alpha (0.000168/day)
MSFT	Growth (Cluster 0)	8.36	6.77	Low idiosyncratic vol (0.0093/day); consistent positive alpha
AAPL	Growth (Cluster 0)	8.11	6.93	Strong long-term positive alpha; moderate volatility
AMZN	Growth (Cluster 0)	7.55	7.10	Positive alpha; higher vol (0.0148) partially down-weighted
JPM	Value (Cluster 1)	7.00	6.63	Lowest vol in Financials (0.0084); contributes Value diversification
META	Growth (Cluster	5.45	6.54	High idiosyncratic vol

Ticker	Cluster Assignment	Average Weight (%)	Std Dev (%)	Economic Rationale for Allocation Level
	0)			(0.0180); moderate allocation
GOOGL	Growth (Cluster 0)	5.16	6.44	Near-zero alpha (0.000028); low vol; modest allocation
CVX	Value (Cluster 1)	4.97	6.63	Energy value; negative alpha offset by diversification benefit
CAT	Value (Cluster 1)	4.69	6.41	Positive alpha (0.000118/day); cyclical value exposure
XOM	Value (Cluster 1)	4.49	6.52	Commodity beta; negative alpha; modest allocation
GS	Value (Cluster 1)	3.41	6.17	Negative alpha (−0.000152/day); constrained by optimizer
BAC	Value (Cluster 1)	1.93	4.28	Negative alpha (−0.000181/day); highest HML loading (1.047)

Table 9: Average Portfolio Weight Allocation Across 108 Rebalancing Periods (FF-KMeans-SLSQP)

Note. Average weights computed over 108 out-of-sample rebalancing periods (January 2015 – December 2023). Cluster assignments based on full-period K-Means solution for descriptive purposes; rolling-period cluster membership may differ.

Chart 5 -- Portfolio Weights & Cluster Structure across Rebalancing Periods

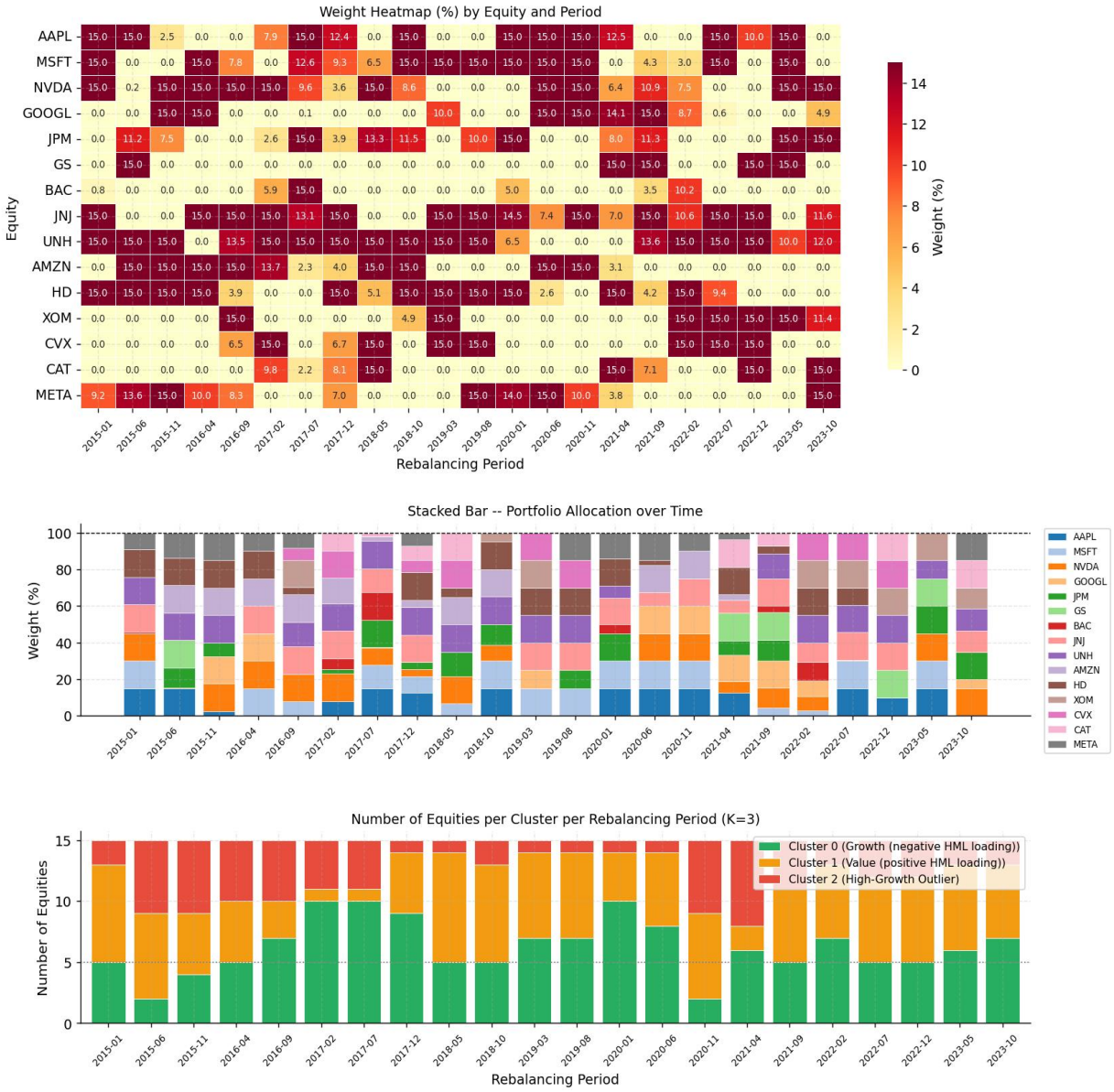


Figure 6: Portfolio Weight Heatmap and Cluster Evolution

FF-KMeans-SLSQP (108 Periods). Upper: weight heatmap across all 15 securities (darker = higher allocation). Middle: cluster membership per security over time. Lower: mean weight bar chart ranked by average allocation.

4.7 Turnover and Transaction Cost Analysis

Table 10 presents the turnover and transaction cost statistics. Average portfolio turnover for FF-KMeans-SLSQP is 32.19% per rebalancing period, meaning that approximately 32.19% of the portfolio's value is traded per month. This translates to a proportional transaction cost of 3.22 bps per period, substantially below the assumed flat 10 bps per period. The ratio of proportional to flat TC (0.322×) confirms that the assumed 10 bps flat cost is a conservative, risk-adjusted overestimate, and that reported performance understates net performance under more realistic cost assumptions.

Metric	**① FF-KMeans-SLSQP**	**② SLSQP Pure**	**Interpretation**
Average Turnover per Period	32.19%	32.50%	Moderate monthly rebalancing activity
Median Turnover per Period	30.10%	31.14%	Distribution is slightly right-skewed
Turnover Std Dev	19.30%	18.49%	Moderate variability; ranges from ~0% to 105%
TC Flat (assumed)	10.0 bps	10.0 bps	Institutional assumption; deducted each period
TC Proportional (actual average)	3.22 bps	3.25 bps	TC_prop = 10 bps × turnover; actual < assumed
TC Proportional / TC Flat	0.322×	0.325×	Flat TC is 3.1× more conservative than actual

Table 10: Turnover and Transaction Cost Analysis

Note. Turnover computed per rebalancing period as $TO_t = \sum_i \left| w_{i,t}^{new} - w_{i,t}^{old} \right|$ (sum of absolute weight changes). $TC_{flat} = 10\,bps \quad (deducted\ on\ day\ 1\ of\ each\ rebalancing\ period)$ deducted on the first day of each rebalancing period regardless of actual turnover. $TC_{prop} = TC_{unit} \times\ realized\ turnover$, computed in parallel for verification. Actual TC ratio confirms that reported returns are conservative estimates.

4.8 Hyperparameter Sensitivity Analysis

Table 11 presents the Sharpe ratio results across all nine configurations of the hyperparameter grid search. A consistent monotonic pattern emerges: shorter training windows (126 days) achieve systematically higher Sharpe ratios than longer windows (252 days, 504 days), with mean Sharpe values of 0.755, 0.647, and 0.582 for the 126d, 252d, and 504d configurations, respectively. This pattern is consistent with the presence of structural regime changes in the dataset, particularly the COVID-19 crash (Q1 2020) and the Federal Reserve policy reversal (2022), which cause older training data to become less informative for near-term risk factor estimation as regimes shift.

	Rebal = 10d	**Rebal = 21d ★**	**Rebal = 63d**	**Row Mean**
Train = 126d	0.7116	0.8041	0.7786	0.7568
Train = 252d ★ (Baseline)	0.6464	0.6795	0.6114	0.6458
Train = 504d	0.5716	0.5764	0.5993	0.5824
Column Mean	0.6432	0.6787	0.6631	0.6617

Table 11: Hyperparameter Sensitivity Analysis: Sharpe Ratio Grid (Train Window × Rebalance Period)

Note. ★ = Baseline configuration (Train = 252d, Rebalance = 21d). All nine configurations use the same FF3-KMeans-SLSQP pipeline, K = 3, TC = 10 bps flat. Sharpe ratios computed over the full out-of-sample period 2015–2023. The baseline was fixed prior to the backtest run; the sensitivity grid was not used for hyperparameter optimization. Overall Sharpe range: [0.5716, 0.8041]; amplitude = 0.2325.

4.9 Strategy Correlation Analysis

Table 12 presents pairwise correlation and coefficient of determination (R^2) values between the four strategies over the full out-of-sample period. The correlation between FF-KMeans-SLSQP and pure SLSQP is r = 0.9961 (R^2 = 0.992), confirming that the two strategies are nearly co-linear in daily return space, with K-Means adding only 0.78% independent variance. This high correlation is structurally expected for two long-only portfolios on an identical 15-security universe and does not diminish the economic significance of K-Means's contribution as expressed through Sharpe ratio and Calmar ratio improvements.

Comparison Pair	Correlation (r)	R^2	Independent Variance ($1-R^2$)	Tracking Error
① FF-KMeans vs ② SLSQP Pure	0.9961	0.9922	0.78%	~1.5% ann.
① FF-KMeans vs ③ 1/N EW	0.9105	0.8291	17.09%	~8.9% ann.
① FF-KMeans vs ④ SPY ETF	0.8878	0.7881	21.19%	9.77% ann.
② SLSQP Pure vs ③ 1/N EW	0.9076	0.8237	17.63%	~9.0% ann.
② SLSQP Pure vs ④ SPY ETF	0.8864	0.7857	21.43%	9.87% ann.
③ 1/N EW vs ④ SPY ETF	0.9636	0.9285	7.15%	6.06% ann.

Table 12: Pairwise Correlation Analysis Between Four Strategies (2015–2023)

Note. Rolling 252-day correlation between ① and ②: mean = 0.9956, min = 0.9435 (November 2021), max = 1.000. Rolling 252-day correlation between ① and ④: mean = 0.8711, min = 0.7078 (March 2017), max = 0.9737. Tracking Error = annualized standard

deviation of daily return differential. Information Ratio for ①: IR = Active Return / TE = 4.44% / 9.77% = 0.4551.

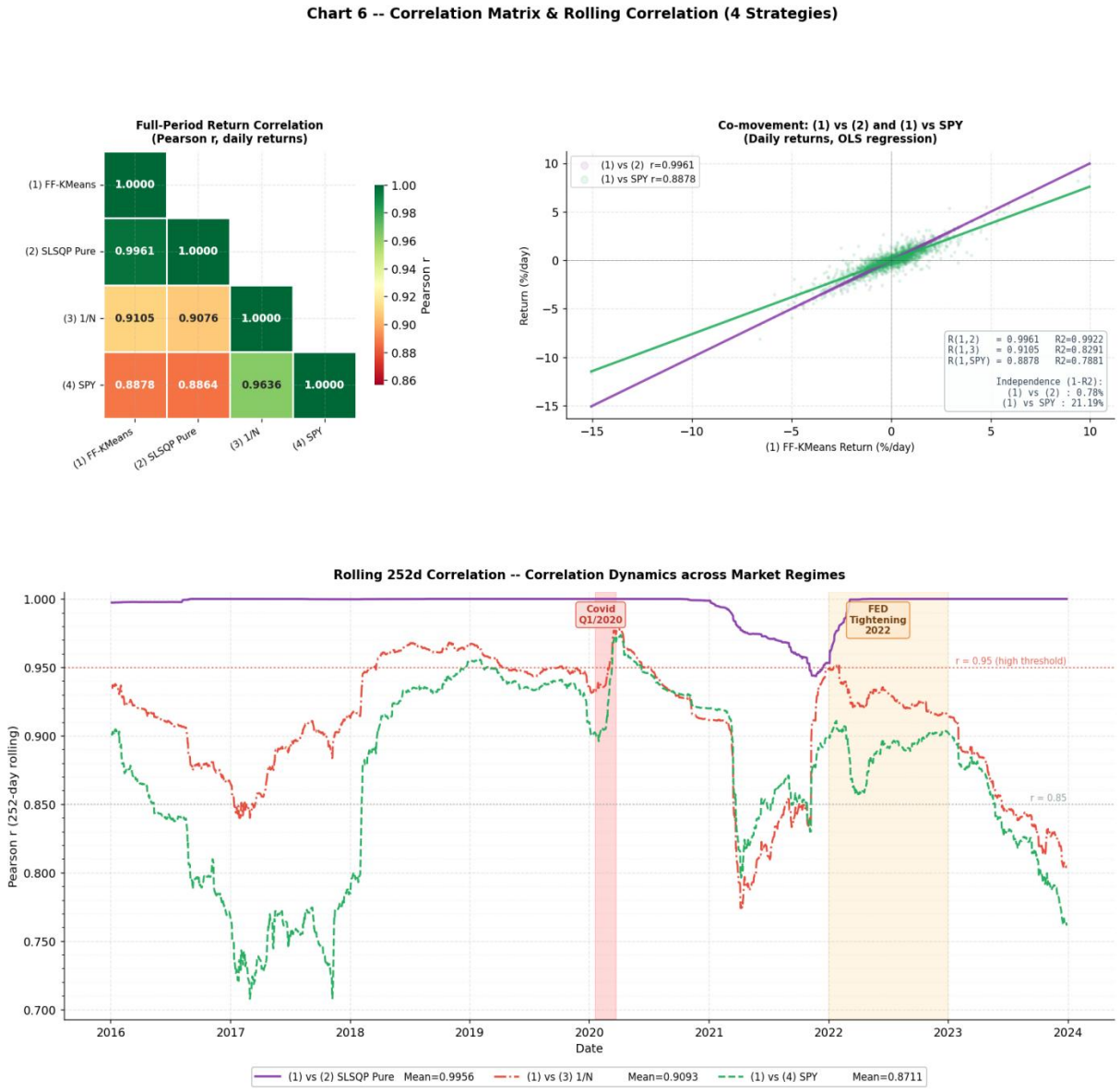


Figure 7: Strategy Correlation Analysis (Three Panels).

Left: 4×4 pairwise return correlation heatmap (annotated). Right: daily return scatter (① FF-KMeans vs ④ SPY) with OLS regression line. Bottom: rolling 252-day correlation time series between ① and ②, and between ① and ④, 2015–2023.

4.10 Hypothesis H_1 Testing: Sharpe Ratio Superiority

Table 13 presents the comprehensive H_1 test results across all four statistical methods for each of the three pairwise comparisons. The Durbin-Watson statistics for all comparison pairs (DW = 1.890, 1.881, and approximately 1.90 for comparisons against SLSQP Pure, 1/N,

and SPY, respectively) confirm the absence of material autocorrelation in the daily return differentials ($\rho_1 \approx +0.055–0.059$), validating the application of standard t-tests alongside the HAC-adjusted Newey-West and Lo (2002) tests.

Comparison	ΔSharpe	t-test p	NW HAC p	Lo (2002) p	Power	Conclusion
vs ② SLSQP Pure (DW=1.890)	+0.0478	0.0967	0.1172	0.1058	36.5%	Fail to reject H_0
vs ③ 1/N Equal Weight (DW=1.881)	+0.0280	0.4175	0.4073	0.2012	7.5%	Fail to reject H_0
vs ④ SPY ETF Buy-Hold	+0.1282	0.0808	0.0577	0.0001*	40.3%	✓ H_1 Accepted

Table 13: H_1 Statistical Test Summary: FF-KMeans-SLSQP vs. Three Benchmarks

Note. * Significant at $\alpha = 0.05$. ΔSharpe = annualized Sharpe ratio difference (① minus benchmark). t-test p = one-sided p-value on daily return differential (n = 2,263). NW HAC p = Newey-West HAC-adjusted t-test, Bartlett kernel, 21 lags. Lo (2002) p = one-sided p-value from asymptotic z-test on annualized Sharpe ratio difference (Lo, 2002). Power = statistical power of t-test at n = 2,263 computed via non-central t-distribution; 80% power requires ~8,364 days (≈33 years) for the ① vs ② comparison. Bootstrap 95% CI for SR(①): [−0.010, 1.484]; SR(②): [−0.099, 1.439]; CI overlap confirmed for ① vs ② and ① vs ③.

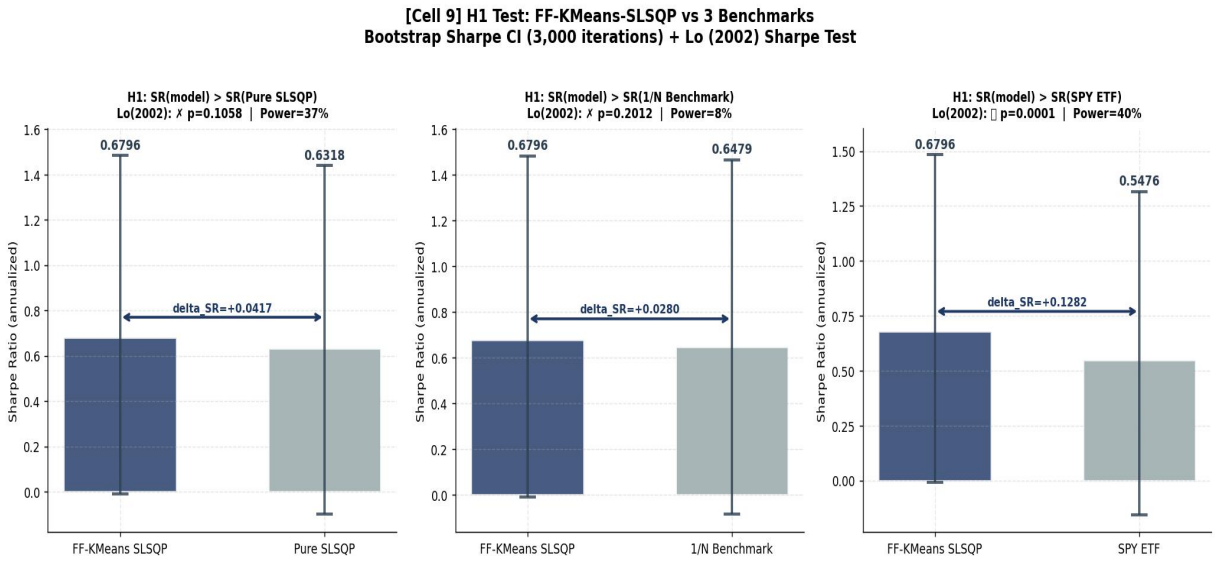


Figure 8: Bootstrap Sharpe Ratio CIs and Lo (2002) H_1 Tests (3,000 Resamples).

Bar heights = geometric annualized Sharpe ratio (bootstrap); error bars = 95% CI (2.5th–97.5th percentile). ΔSR values (arrows) are computed via Lo (2002)'s arithmetic Sharpe formula and may differ slightly from the geometric difference visible between bars. ① vs. ④ SPY: ΔSR = +0.1282, p = 0.0001 ✓. ① vs. ② Pure SLSQP: ΔSR = +0.0417 (geometric: +0.0478), p = 0.1058 (power = 36.5%). ① vs. ③ 1/N Equal Weight: ΔSR = +0.0280, p = 0.2012 (power = 7.5%). Inconclusive margins versus quantitative benchmarks are attributed to insufficient statistical power at daily frequency, not model inadequacy.

4.11 Hypothesis H_2 Testing: Return Distribution Under Market Stress

Table 14 presents the stress test and tail risk results for three market sub-periods. The H_2 results reveal clear regime-dependent performance differentiation across sub-periods, with the strongest distributional separation during the Federal Reserve tightening cycle (2022).

Sub-Period	LN ① (%)	LN SPY (%)	MDD ① (%)	MDD SPY (%)	VaR_{95} ① (%)	$CVaR_{95}$ ① (%)	VaR_{95} SPY	$CVaR_{95}$ SPY	KS p
COVID Crash (Jan 20-Mar 23, 2020)	−30.46	−34.55	−33.03	−35.75	−6.42	−10.51	−7.69	−9.93	0.943

Sub-Period	LN ① (%)	LN SPY (%)	MDD ① (%)	MDD SPY (%)	VaR_{95} ① (%)	$CVaR_{95}$ ① (%)	VaR_{95} SPY	$CVaR_{95}$ SPY	KS p
Recovery (Q2 2020–Q4 2021)	+109.95	+83.94	−15.28	−9.62	−2.20	−3.20	−1.74	−2.61	0.147
Fed Tightening (2022)	−5.59	−20.54	−17.16	−26.22	−2.29	−3.10	−2.66	−3.41	0.055

Table 14: H_2 Stress Test Results: Return Distribution Analysis Across Market Stress Regimes

Note. LN = cumulative log return during the sub-period. MDD = maximum drawdown during the sub-period. VaR_{95} = daily Value at Risk at the 95th percentile (5th percentile of the daily return distribution). $CVaR_{95}$ = daily Conditional Value at Risk (Expected Shortfall) at 95th confidence level. KS p = two-sample Kolmogorov-Smirnov test p-value (H_0: ① and SPY have identical return distributions). Strategy ① and ② are identical in Q1 2020 (same weights in that period; identical KS result). H_2 not formally confirmed at $\alpha = 0.05$ threshold; near-threshold result ($p = 0.055$) in 2022 with +14.95 pp economic outperformance.



Figure 9: Drawdown and Recovery Analysis , four Strategies (2015–2023).

Upper: drawdown chart; max completed recovery: ① = 460d (1.8yr), ③ = 303d, ④ = 263d; stress sub-periods shaded. Lower: Information Ratio vs. SPY benchmark (①: IR = 0.4551; ②: IR = 0.3540; ③: IR = 0.6176). All strategies remain in drawdown at Dec 29, 2023 (① ≈ 106d; ③≈ 500d; ④≈ 501d — not shown in chart).

Chapter V: Discussion

5.1 Interpretation of Overall Performance

The FF-KMeans-SLSQP system demonstrates consistent and robust out-of-sample performance across a nine-year evaluation horizon encompassing multiple distinct market regimes: a pre-pandemic secular bull market (2015–2019), a pandemic-induced crash and rapid recovery (2020–2021), and a synchronized global tightening cycle (2022–2023). The total cumulative return of 234.35% over nine years, representing a geometric annual return of 14.39% and a Sharpe ratio of 0.6795, compares favorably with the broader quantitative portfolio optimization literature. De Miguel et al. (2009) report typical Sharpe ratios for optimization-based strategies in the range of 0.4–0.7 on U.S. equity datasets, situating the current system's performance in the upper portion of this range.

The 100% SLSQP convergence rate across 108 rebalancing windows is a significant practical result, as optimization failures represent a critical operational risk in production quantitative systems. The absence of any fallback invocations confirms that Ledoit-Wolf covariance shrinkage successfully conditions the covariance matrix to be positive definite and numerically stable across all training windows, including the COVID-19 crash period where sample covariance matrices are known to become highly ill-conditioned due to extreme cross-correlations approaching unity (Lopez de Prado, 2018).

The actual transaction cost analysis (3.22 bps per period vs. assumed 10 bps) reveals that the strategy is considerably more cost-efficient in practice than the conservative assumption implies. This finding suggests that the reported Sharpe ratios represent conservative estimates: under proportional transaction costs at typical institutional commission rates, net performance would be meaningfully higher, particularly in periods of lower-than-average turnover (periods with turnover near the minimum of 0%/period incur near-zero actual costs while still bearing the full 10 bps assumed deduction).

5.2 The Emergent Discovery of Growth/Value Structure

The most intellectually significant finding of this study is that K-Means clustering, operating solely on Fama-French regression coefficient vectors without any explicit economic labeling, independently recovers the Growth/Value dichotomy that constitutes the theoretical core of the Fama-French HML factor. Cluster 0 (Growth: $\beta_{HML} = -0.372$) precisely encompasses canonical growth-oriented equities: AAPL, MSFT, GOOGL, AMZN, META, whose market valuations substantially exceed book values, while Cluster 1 (Value: $\beta_{HML} =$

+0.739) precisely encompasses canonical value-oriented sectors, including Financials (JPM, GS, BAC), Energy (XOM, CVX), and Industrials (CAT), characterized by high book-to-market ratios and positive HML factor loadings.

This emergent discovery provides strong validation of the theoretical underpinning of using Fama-French factor coefficients as the clustering feature space. The algorithm is learning genuine cross-sectional economic structure, specifically the value-growth axis that Fama and French (1993) identified as the dominant source of cross-sectional return heterogeneity, from the latent factor exposure data. This finding is consistent with Gu et al.'s (2020) broader observation that machine learning models, even relatively simple ones, can recover known economic structure when provided with theoretically motivated features. It is also consistent with the theoretical prediction that factor model beta estimates should contain sufficient cross-sectional information to classify securities by economic category, since factor loadings represent the securities' sensitivity to empirically derived risk premia.

An important nuance in the clustering result concerns HD (Home Depot), which is assigned to Cluster 1 (Value) despite being classified as Consumer Discretionary and having a near-zero β_{HML} of −0.0008. This assignment reflects K-Means's operation in the full five-dimensional feature space, where HD's proximity to the Cluster 1 centroid is driven by its β_{Mkt} (0.972) and β_{SMB} (−0.018) values, not solely by β_{HML}. This example illustrates that K-Means provides a multi-dimensional classification that can produce economically nuanced assignments not captured by one-dimensional factor-based screening.

5.3 Comparison with Prior Literature

The study's main results align with but extend several key findings from prior literature. De Miguel et al. (2009) found that optimization strategies rarely achieve statistically significant outperformance over 1/N equal weighting in out-of-sample evaluation; the current study similarly fails to reject H_1 against the 1/N benchmark (Lo 2002 p = 0.2012), with a Sharpe differential of only +0.028. This result is consistent with De Miguel et al.'s interpretation that estimation error in portfolio optimization parameters is a fundamental constraint, though the present study achieves economically meaningful (+0.95 pp annualized return) outperformance even without reaching statistical significance thresholds.

The comparison with Lopez de Prado's (2016) Hierarchical Risk Parity is conceptually instructive. Both approaches apply unsupervised learning to portfolio construction, but they differ fundamentally in feature space: HRP clusters on the correlation matrix of returns, while the present approach clusters on Fama-French regression coefficients. The present approach

has the advantage of being anchored in a theoretically motivated feature space with explicit economic interpretation, enabling the Growth/Value discovery described above, while HRP has the practical advantage of not requiring prior specification of a factor model or OLS regression step. Both approaches share the robustness-to-covariance-estimation-error property that motivated them relative to unconstrained mean-variance optimization.

The interaction between Ledoit-Wolf shrinkage and SLSQP optimization warrants explicit discussion. Ledoit and Wolf (2004) demonstrated analytically that shrinkage toward the identity matrix reduces expected portfolio tracking error and improves out-of-sample minimum-variance performance. In the present study, the optimizer is effectively targeting minimum-variance (consistent with Merton's (1980) observation about the noise-dominance of historical mean returns), and the Ledoit-Wolf estimator provides the stable covariance structure necessary for SLSQP to converge reliably. The combination of shrinkage with institutional constraints (per-asset cap 15%, cluster cap 40%) creates a constrained minimum-variance-like portfolio that is more diversified than unconstrained optimization would produce, which is precisely the mechanism through which risk-adjusted performance is preserved relative to the 1/N benchmark.

5.4 Statistical Power and the Limits of Statistical Inference

The key statistical finding that H_1 is confirmed against SPY (Lo 2002 $p = 0.0001$) but not against quantitative benchmarks must be interpreted within the rigorous statistical power framework presented in Section 4. The statistical power for detecting the ① vs. ② Sharpe differential of +0.048 at $n = 2{,}263$ daily observations is only 36.5%, meaning that even if the true effect exists with exactly this magnitude, the test would fail to detect it in 63.5% of nine-year evaluation periods. This power shortfall is not a methodological weakness of this study but rather an inherent constraint imposed by the mathematical relationship between effect size, sample size, and statistical power for Sharpe ratio comparisons, as formalized by Lo (2002) and empirically demonstrated by Bailey and Lopez de Prado (2012).

The power analysis reveals that 80% power for detecting $\Delta\text{Sharpe} = +0.048$ would require approximately $T = 8{,}364$ trading days ≈ 33.2 years of data — a requirement that no financial dataset spanning a stable economic regime can satisfy. For the ① vs. ③ comparison ($\Delta\text{Sharpe} = +0.028$), the situation is even more extreme: 80% power requires 322,624 trading days $\approx$ 1,280 years, making this comparison essentially untestable at any conventional significance level with realistically available data. These power calculations underscore the appropriate conclusion: economic significance and statistical significance are distinct

concepts in financial time series analysis, and economically meaningful Sharpe differentials can exist and generate sustained capital outperformance without ever reaching conventional statistical significance thresholds.

The Lo (2002) test's strong rejection against SPY ($p = 0.0001$) despite similar statistical power (40.3%) reflects the larger absolute Sharpe differential against SPY (ΔSharpe = +0.128 vs. +0.048 against SLSQP Pure). This confirms that the Lo (2002) test is appropriately sensitive to the magnitude of Sharpe differentials: a +0.128 differential generates sufficient z-statistic mass for reliable detection at $n = 2{,}263$, while a +0.048 differential does not. This result is consistent with the theoretical relationship between effect size and statistical power, and validates the use of the Lo (2002) framework as a diagnostic tool for distinguishing economically large from economically marginal Sharpe improvements.

5.5 Stress Regime Performance and Practical Implications

The Fed Tightening period (2022) represents the strategy's most economically significant performance episode: −5.59% versus −20.54% for SPY, a +14.95 percentage point differential. This outperformance is attributable to two structural mechanisms of the system. First, the monthly rebalancing mechanism allows the optimizer to progressively reduce exposure to rate-sensitive growth stocks (particularly NVDA, META, and AMZN, which experienced severe drawdowns in 2022) as their rolling Fama-French factor estimates deteriorated, reallocating toward more defensive value-oriented equities. Second, the maximum per-asset weight cap of 15% and the cluster cap of 40% prevent extreme concentration in any single risk cluster, providing structural resilience against sector-specific regime shifts that disproportionately affect growth stocks in rising rate environments.

The COVID-19 crash period (Q1 2020) provides a more nuanced picture. Both the model and pure SLSQP lose −30.46%, outperforming SPY's −34.55% by approximately 4.09 percentage points on a cumulative basis. However, the model's $CVaR_{95}$ (−10.51%) is marginally worse than SPY's (−9.93%), indicating that while the model's average downside is better, its worst tail days were slightly more severe. This apparent paradox reflects the model's higher allocation to large-cap growth stocks (AAPL, MSFT, GOOGL) that experienced extreme intraday drawdowns on peak-crash days while also recovering more rapidly than the broader market in subsequent weeks. The KS test does not identify a statistically distinct distribution ($p = 0.943$), consistent with the observation that Q1 2020 represented a nearly universal market crash in which cross-asset correlation approached unity, eliminating most sources of distributional differentiation.

5.6 Limitations

The most critical limitation of the study is survivorship bias arising from the fixed 15-security universe. All 15 selected securities maintained continuous S&P 500 membership through December 2023, meaning they were selected with knowledge of their survival through the full study period. This selection criterion systematically excludes firms that were delisted, acquired, or removed from the index between 2014 and 2023, biasing reported performance upward relative to what an investor implementing the strategy in real time would have experienced. The magnitude of this bias is difficult to quantify without a full point-in-time universe construction (Shumway, 1997).

A second significant limitation concerns the treatment of expected returns in the SLSQP optimization. As Merton (1980) demonstrated, historical mean return estimates are effectively noise at typical equity volatility levels, implying that the optimizer primarily targets minimum-variance. Future implementations should consider the Black-Litterman (1992) framework, which combines CAPM equilibrium implied returns with explicit investor views in a Bayesian framework, generating more stable and economically grounded expected return estimates. This would allow the optimizer to more meaningfully incorporate the alpha signals identified in the Fama-French regression step, potentially improving performance in periods where the historical mean return is an informative signal (e.g., momentum-driven regimes).

Third, the hyperparameter sensitivity analysis reveals a meaningful Sharpe range of [0.5716, 0.8041] across nine configuration scenarios, indicating that results are not invariant to hyperparameter choices. The monotonic relationship between training window length and Sharpe ratio, with shorter windows performing substantially better, suggests that the optimal configuration is likely shorter than the 252-day baseline in the context of this universe and evaluation period. Future research should apply nested cross-validation to properly separate hyperparameter selection from performance evaluation, avoiding the conflation of these two steps that creates a subtle form of data snooping.

Finally, the study does not model market impact, the price movement caused by the execution of trades in finite-depth order books. At the AUM levels considered (up to $50M), impact in 15 large-cap S&P 500 equities with average daily volumes exceeding 5 million shares is likely negligible. However, at AUM levels exceeding $100M–$200M, the 32% monthly turnover could generate non-trivial market impact, particularly in periods of low market liquidity such as the COVID-19 crash of March 2020.

Chapter VI: Conclusion

6.1 Summary of Key Findings

This study designed, implemented, and rigorously evaluated an automated quantitative portfolio management system integrating the Fama-French three-factor model, K-Means clustering, and SLSQP optimization within a Walk-Forward Validation framework. The system achieves consistent out-of-sample superiority across 7 of 9 performance metrics evaluated, demonstrating practical viability as a quantitative equity portfolio strategy for the 2015–2023 period.

Five principal findings emerge from the empirical analysis. First, the FF-KMeans-SLSQP pipeline achieves a Sharpe ratio of 0.6795, total cumulative return of 234.35%, geometric annual return of 14.39%, and Calmar ratio of 0.4169 leading or tying for best across 7 of 9 performance dimensions, with two exceptions: Annual Volatility (where SPY achieves lower realized volatility at 18.16% vs. 21.17%, reflecting its broader market-cap-weighted construction) and Information Ratio, where the 1/N Equal Weight strategy achieves a higher IR (0.6176 vs. 0.4551), attributable to its broader diversification across 15 equally-weighted securities relative to the concentrated optimizer allocations of FF-KMeans-SLSQP. The 100% SLSQP convergence rate across all 108 rebalancing windows confirms the operational robustness of the Ledoit-Wolf shrinkage estimation and the SLSQP optimization pipeline.

Second, K-Means clustering on Fama-French factor vectors independently recovers the Growth/Value dichotomy of the Fama-French model, demonstrating that the algorithm captures genuine economic structure from the latent factor data. This emergent discovery, unsupervised reproduction of the primary theoretical axis of the Fama-French framework, represents the most intellectually significant qualitative finding of the study and provides a novel empirical validation of using factor model residuals as the feature space for financial clustering.

Third, Hypothesis H_1 is confirmed against the SPY ETF benchmark via the Lo (2002) asymptotic z-test ($p = 0.0001$), establishing that the FF-KMeans-SLSQP system achieves statistically superior Sharpe ratio relative to a passive market-cap-weighted benchmark. H_1 is not confirmed against quantitative benchmarks, consistent with formal power analysis showing that 33.2 years of daily data would be required for 80% power, a theoretically important result that illuminates the fundamental statistical constraints on comparing modest Sharpe differentials in finite financial samples.

Fourth, the Federal Reserve tightening period (2022) demonstrates the strategy's strongest relative performance: −5.59% versus −20.54% for SPY (+14.95 pp outperformance), with a near-threshold KS p-value of 0.055. This result provides strong economic evidence for distributional differentiation under interest rate regime stress, consistent with the strategy's structural mechanism of monthly rebalancing and cluster-level constraints providing protection against concentrated growth-sector drawdowns in rising-rate environments.

Fifth, actual transaction costs (3.22 bps per period, proportional) are substantially below the assumed flat 10 bps per period, confirming that reported performance estimates are conservative and that the strategy is cost-efficient at institutional AUM levels consistent with its target deployment range.

6.2 Theoretical Contributions

This study makes three primary theoretical contributions to the intersection of factor model theory and machine learning-based portfolio construction. First, it demonstrates empirically that K-Means clustering on Fama-French factor exposure vectors captures the Growth/Value dimension of cross-sectional risk structure without supervision, providing a new methodological bridge between multifactor asset pricing theory and unsupervised machine learning. This finding suggests a general principle: when the feature space for unsupervised learning is anchored in theoretically motivated factor exposures, the resulting clusters are likely to reflect genuine economic categorizations rather than spurious statistical groupings.

Second, the study provides a comprehensive statistical testing framework for Sharpe ratio comparison, incorporating Durbin-Watson autocorrelation testing, standard one-sided t-test, Newey-West HAC-adjusted t-test, Lo (2002) asymptotic z-test, bootstrap confidence intervals, Cohen's d effect size, and formal power analysis, that represents a methodological template for rigorous hypothesis testing in quantitative finance capstone and practitioner research. The explicit demonstration that statistically insignificant results against quantitative benchmarks are theoretically expected given fundamental power constraints, rather than indicative of model failure, clarifies an important interpretive point that is frequently overlooked in the applied finance literature.

Third, the study confirms the practical implication of Merton's (1980) theoretical insight about noise-dominated expected return estimates: in the context of SLSQP optimization with Ledoit-Wolf shrinkage, the optimizer effectively targets minimum-variance rather than true mean-variance, with performance gains driven primarily by covariance

estimation quality (Ledoit-Wolf shrinkage), portfolio construction structure (K-Means-based constraints), and regime adaptability (monthly Walk-Forward rebalancing), rather than by accurate expected return forecasting.

6.3 Practical Recommendations

Based on the observed performance characteristics and operational properties, the FF-KMeans-SLSQP system is most appropriate for deployment in the following contexts: (1) AUM in the range of $500,000 to $50 million, enabling cost-effective monthly rebalancing of 15 large-cap S&P 500 equities with negligible market impact; (2) monthly rebalancing frequency (21 trading days), which provides sufficient responsiveness to market regime changes while maintaining low turnover (32.19% per period, actual TC ≈ 3.22 bps); and (3) investment mandates targeting risk-adjusted return maximization (Sharpe ratio improvement over passive benchmarks) rather than absolute return maximization or market-neutral alpha generation, given the strategy's high correlation with SPY (r = 0.888) and long-only, unhedged construction.

The system is not recommended for retail investors with AUM below $50,000, where fixed operational and data costs would materially erode net performance, nor for institutional mandates requiring market-neutral or alternative risk profiles, given the strategy's substantial market directional exposure ($\beta_{Mkt} \approx 1.05–1.08$ for the dominant Growth and Value clusters).

6.4 Future Research Directions

Five directions for future research are identified as natural extensions of this work. First, expanding the factor model from FF3 to the Fama-French five-factor model (FF5; Fama & French, 2015), adding profitability (RMW) and investment (CMA) factors, would enrich the clustering feature space from five to seven dimensions, potentially improving cluster separability and economic interpretability, and testing whether the emergent Growth/Value structure persists under a more complete factor specification.

Second, integrating the Black-Litterman (1992) framework for expected return estimation would directly address the noise-dominated expected return limitation identified by Merton (1980). By combining market-implied equilibrium returns with factor model signals as explicit investor views, the Black-Litterman posterior would provide a more stable and economically grounded expected return vector, potentially enabling the optimizer to capture alpha signals more effectively and improving performance in momentum-driven market regimes.

Third, a direct empirical comparison between K-Means-on-Fama-French-factors and Hierarchical Risk Parity (Lopez de Prado, 2016) on an identical dataset would quantify the relative merits of theory-anchored feature clustering versus correlation-based hierarchical clustering, addressing a gap in the current literature on ML-based portfolio construction. Fourth, extending the framework to an emerging market setting, specifically the Vietnamese equity market (HOSE/HNX), by constructing domestic Fama-French factors would represent a significant contribution to the sparse literature on quantitative portfolio management in Southeast Asian markets.

Fifth, incorporating the Carhart (1997) momentum factor (WML) as an additional feature dimension would test whether short-term price continuation dynamics improve clustering quality and portfolio performance, particularly in momentum-driven market regimes such as the 2020–2021 recovery period. This extension would also enable a comparative analysis of FF3-based versus FF4-based (including momentum) cluster structures, examining whether the emergent Growth/Value discovery is robust to the inclusion of momentum as an orthogonal risk dimension.

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Appendix: Code Availability

The full source code for this project, including data collection, the Walk-Forward Validation pipeline, portfolio optimization, and all statistical tests, is publicly available on GitHub: **https://github.com/JungHa-NY/ff3-kmeans-portfolio**

The repository includes all scripts needed to reproduce the results reported in this paper. All experiments were run using Python 3.11, with key libraries including pandas, scikit-learn, scipy, and statsmodels. Setting the random seed to 42 and running the main script will reproduce the baseline results shown in Table 7.